HOMEWORK 4.5 · UNIT 4 · NC.8.G.5

Triangle Angle Relationships

Targets: G.5-3, G.5-4, G.5-5, G.5-2, G.5-6, G.5-7, G.5-8, G.5-9

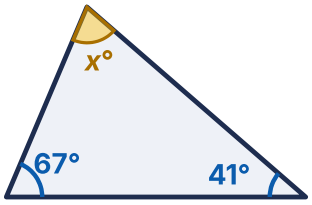
Name _____ Date _____ Block _____

Remember: Triangle Sum: the 3 angles add to 180°. **Exterior Angle:** an outside angle = the sum of the 2 far (remote) inside angles. **AA:** if 2 angles of one triangle match 2 angles of another, the triangles are similar. Example: 50 + 60 + x = 180 → x = 70.

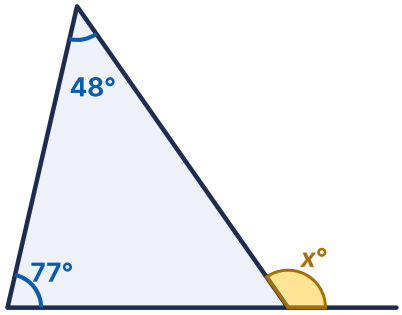
SCORE 3.0 Approaching · the basics

Targets: G.5-3, G.5-4, G.5-5, G.5-2

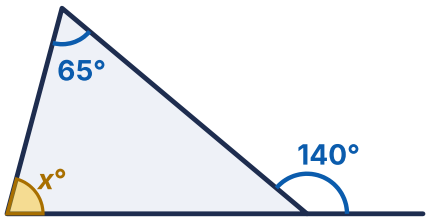
1 Find x. G.5-3



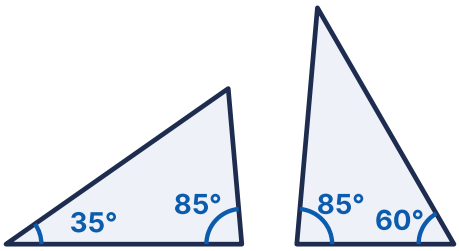
2 Find the exterior angle x. G.5-4



3 Find x. G.5-4



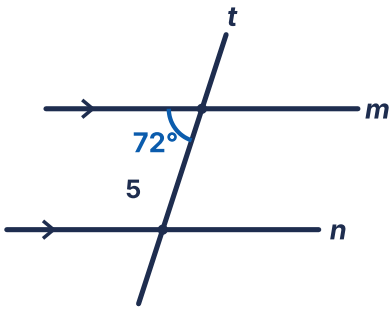
4 Are the two triangles similar? G.5-5



Yes No

Why? _____

5 $m \parallel n$ and $\angle 3 = 72°$. What is $\angle 5$? REVIEW 4.3 G.5-2

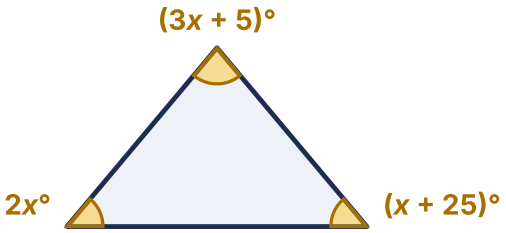


Ⓐ 72° Ⓑ 108° Ⓒ 18° Ⓓ 288°

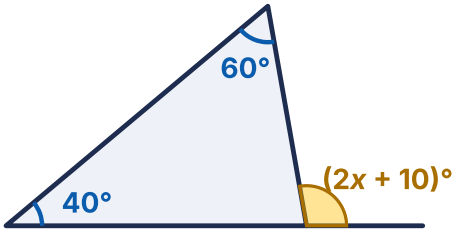
SCORE 4.0 Proficient · on target

Targets: G.5-6, G.5-7

6 Find x. Then find all three angles. G.5-6



7 **Grace** made a mistake. Circle it. Explain it. Fix it. G.5-7



GRACE'S WORK

$2x + 10 + 40 + 60 = 180$

$2x + 110 = 180$

$2x = 70$, so $x = 35$

8 ∠A and ∠B are complementary. ∠A = $(4x + 2)°$ and ∠B = $(x + 3)°$. What is x? REVIEW 4.1 G.5-6

Ⓐ 17 Ⓑ 35 Ⓒ 19 Ⓓ 85

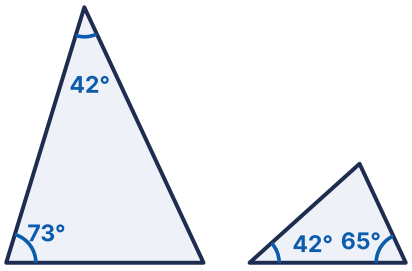
SCORE 5.0 Mastery · go deeper

Targets: G.5-8, G.5-9

9 Can a triangle have an exterior angle of 40° and a remote interior angle of 50°? Explain. G.5-8

_____ , because the exterior angle _____ .

10 Triangle 1 has angles of 42° and 73°. Triangle 2 has angles of 42° and 65°. Are they similar? Show your work. G.5-9



The third angle is _____ . So _____ .